

# OpenWebSearch.eu — A European Open Web Index and its potential Applications

Prof. Dr. Michael Granitzer Chair of Data Science, University of Passau Coordinator  
OpenWebSearch.eu

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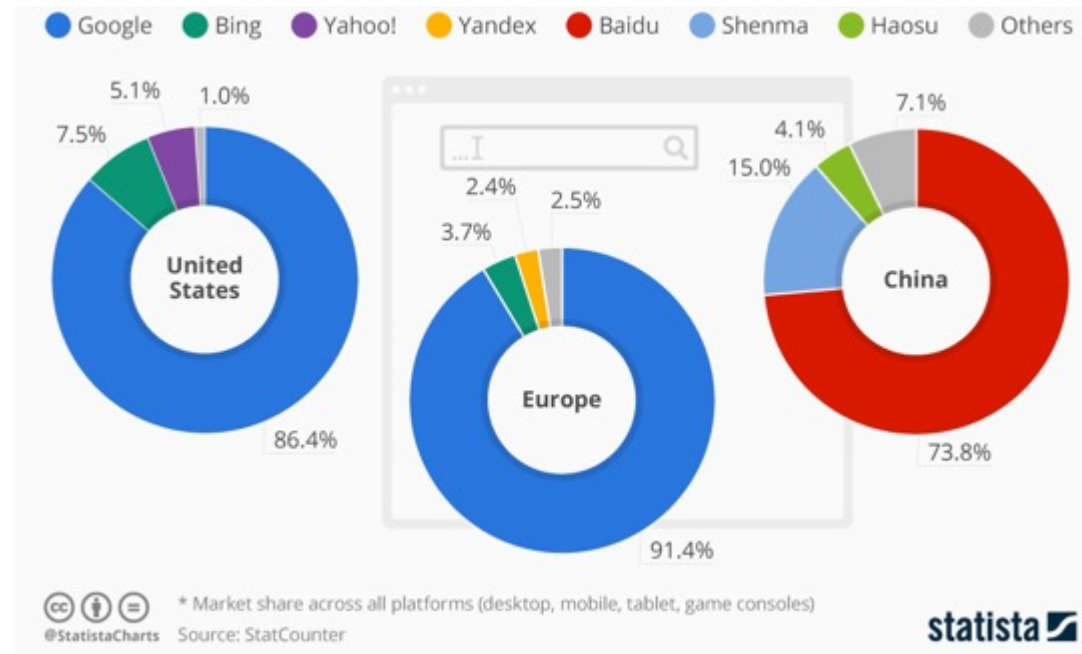


# Web Search Today

A critical resource managed as an oligopoly

## Two properties that don't fit

- A **critical infrastructure** for society, comparable to satellite navigation
- A **market oligopoly**: dominated by a small number of gatekeepers (Google, Microsoft, Baidu, ...)



# But does it matter? Yes — on four dimensions

## User Choice & Democracy

- Single gatekeeper for society's information
- Search rankings can shift voting preferences by **20–72%** (SEME)
- Monopolistic ranking = single point of failure for information quality

## Untapped Richest Information Source

- 60 % of web pages carry structured data
- No open access to web-scale document collections
- Science, research and technology heavily depend on the Web as platform

## Societal & Security Risks

- European OSINT depends on US-controlled infrastructure
- Filter bubbles, algorithmic bias
- Misinformation & Disinformation

## AI & Innovation

- Web data is the fuel for LLMs, RAG, and agentic AI
- Closed indices block reproducible research & open innovation

Being able to navigate the Web is as fundamental as GPS or the power grid

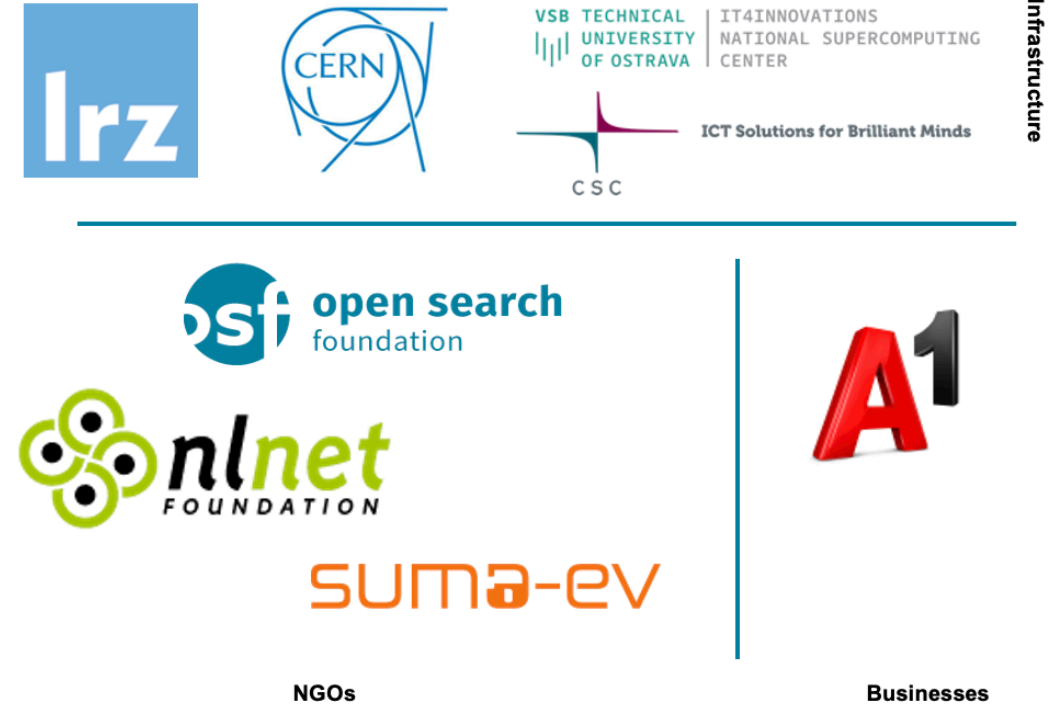
# OpenWebSearch.eu — Our Mission

Building an Open Index of the Web and an open infrastructure to reduce upfront costs for researchers, innovators and companies in tapping the Web as a resource for search, web-analytics and AI.

## Four Objectives

- 1. Open Technology Stack**  
Open-source pipelines for crawling, preprocessing, indexing
- 2. Resource Provision** Building a network of infrastructure providers (EuroHPC / AI Factories)
- 3. Added Value Services**  
Dashboard, tools, data access APIs
- 4. Bootstrapping the Ecosystem** Third-party calls, community building, applications and use-cases

## 14 Core Partners

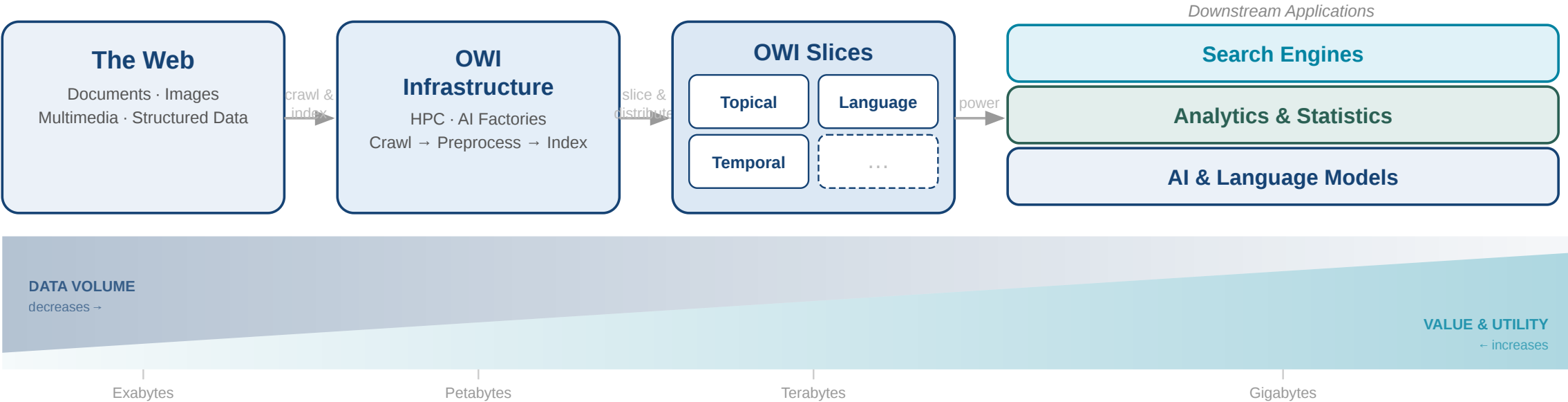


Plus 16 financially supported third-party projects (FSTPss)

# The Open Web Index (OWI)

A **Web Index** = a data structure for fast query-based access and ranking of web documents — the core of every web search engine

**Our Proposition:** A collaboratively created, open and transparent Web Index running on existing HPC computing centres / AI Factories in Europe



Granitzer, Michael, et al. "Impact and development of an Open Web Index for open web search." *Journal of the Association for Information Science and Technology* (2023).

# What is the Open Web Index?

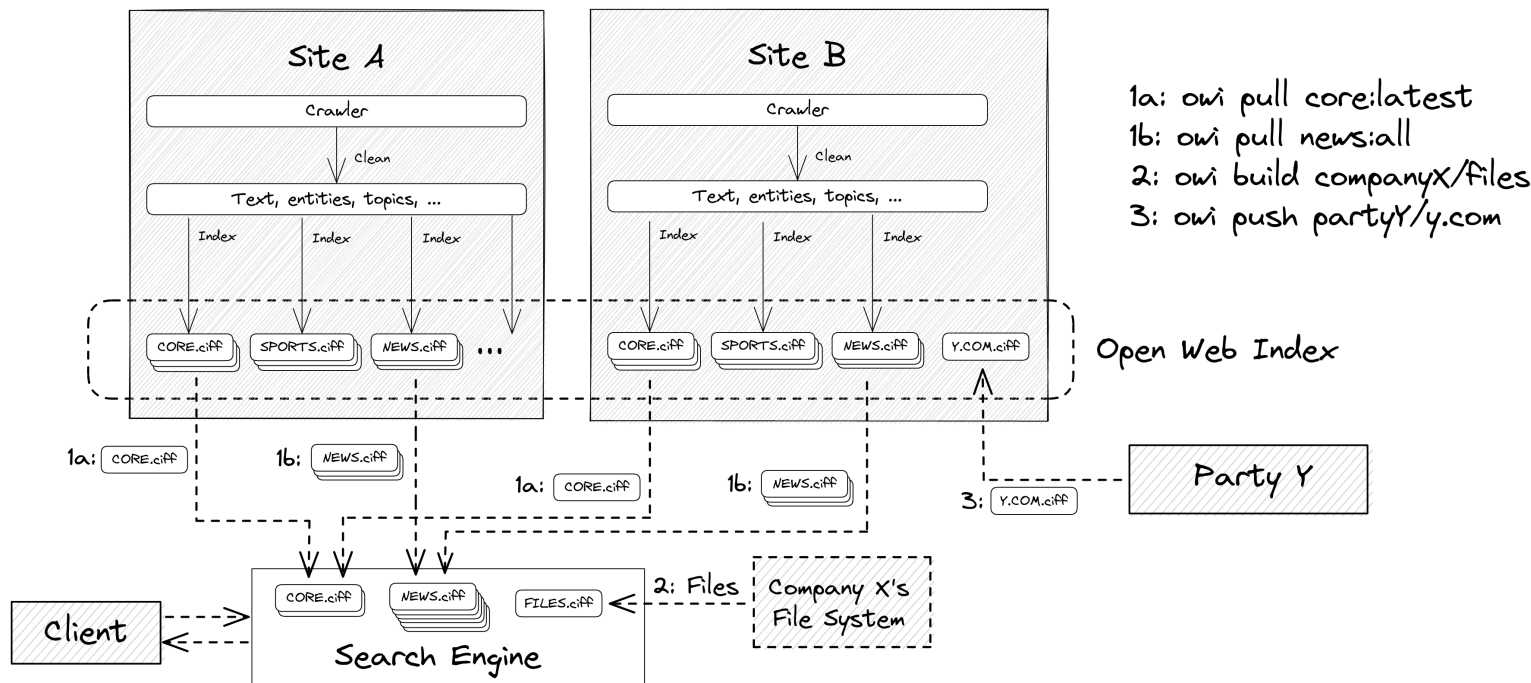


# The Open Web Index

**The Open Web Index** — a federated, daily-published web index built across European HPC sites and consumed as composable shards.

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## Federated production

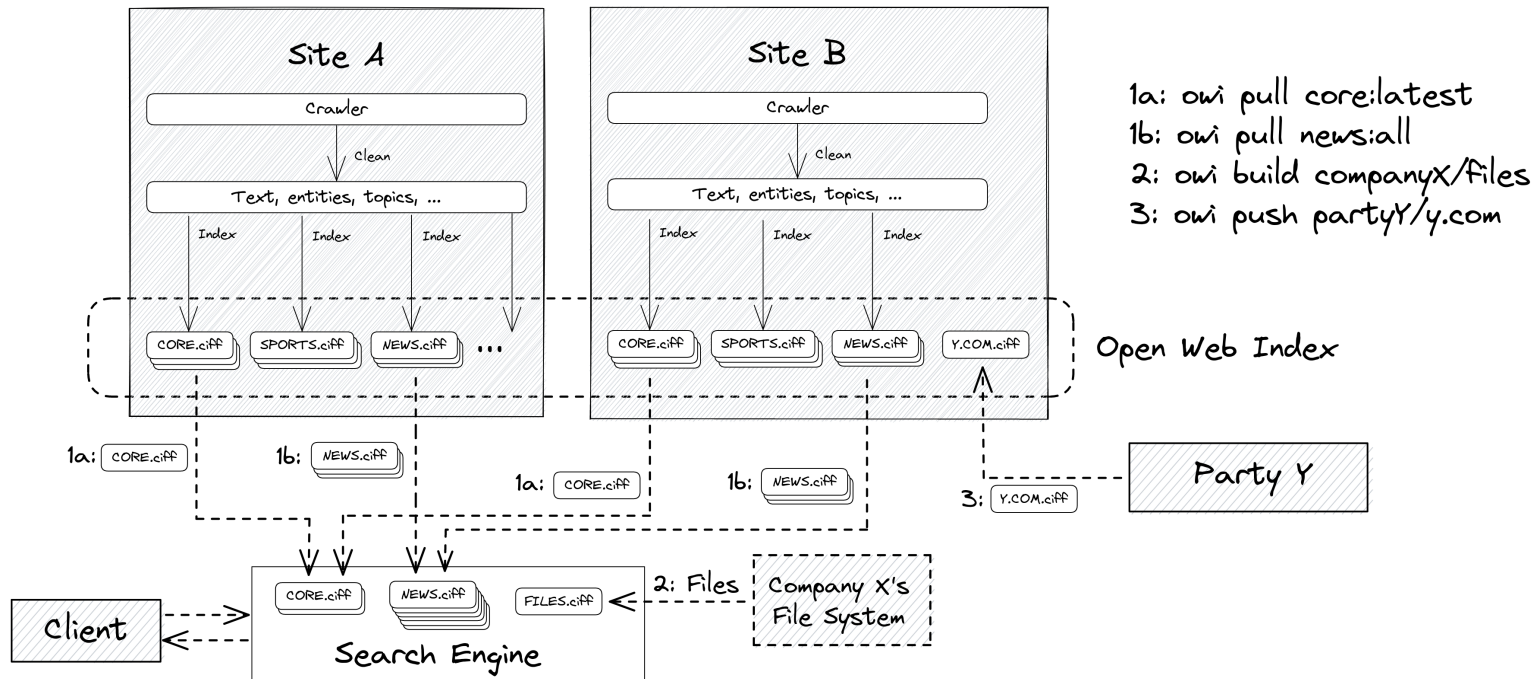
- each HPC site runs the same pipeline independently
- daily shards by date, language, topic

## Three access verbs

- owi pull — download pre-built shards
- owi build — index your own data
- owi push — contribute shards back

# The Open Web Index

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Three output types per daily run — the building blocks of a shard:

# OWI at a Glance — Statistics (Feb 2026)

## Crawling

|                 |                           |
|-----------------|---------------------------|
|                 |                           |
| Pages crawled   | ~39 B total, ~12 B unique |
| Unique hosts    | ~500 M (extrapolated)     |
| Crawling effort | 1,927 machine-days        |
| Avg. throughput | ~20 M URLs / day          |

## Index

|                   |                            |
|-------------------|----------------------------|
|                   |                            |
| Documents indexed | ~10.7 B                    |
| Unique domains    | ~75 M                      |
| Public datasets   | 511 main / 1,511 total     |
| Index size        | 45 TB (main) / 50 TB total |
| Embeddings        | 119 M docs · 884 GB        |

- Top languages:** English (38.9%), German (7.2%), Spanish (6.4%), French (5.7%), Russian (5.2%)
- Multimedia Objects:** Roughly 2-3 quality images and 0.5 Videos per Web-Document
- Comparison:** To meet commercial index size, OWI would need to scale by a factor of 20× — factor 10–50× for multimedia

# Step 1: Crawling the Web



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## OWLer — Open Web Crawler

- Up to **105 M URLs / day** across 4 European data centers
- Central URL Frontier at **CERN** — 2 B+ URLs in crawl space
- **1.25 PiB** collected · **1,927 machine-days** · ~39.6 B fetches
- European language and domain bias
- Only HTML; No headless browsing

## Constraints

- Downstream processing caps: **1–2 TB/day per DC**
- Factor ~20× below commercial freshness
- Requires 2–4 dedicated engineers for 24/7 ops



# Step 2: Preprocessing

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## Resilipipe — WARC → Parquet Pipeline

- Two-stage pipeline: **Resilipipe (CPU)** for extraction & enrichment, **Resilipipe GPU** for embeddings & classifiers
- Modular architecture — 10+ pluggable modules (links, structured data, quality scoring, trigger warnings, ...)
- Custom parallelization on HPC: SLURM + Apache Airflow orchestration
- Output: column-oriented **Apache Parquet** with per-document metadata, embeddings, and prediction scores

## Scale & Constraints

- **108 M documents** embedded with jina-embeddings-v3 (785 GB across 55 published datasets)
- GPU stage bottleneck: embedding throughput limits end-to-end pipeline speed
- Downstream cap of **1–2 TB/day per data center** inherited from crawling

## Key Enrichments

- Content extraction & language detection
- Link graph & structured data (FAQs, geo, licenses)
- Curlie taxonomy labels
- Quality scoring (ONNX)
- Trigger warnings (14 languages, 19 categories)
- Dense embeddings (jina-embeddings-v3)
- Phishing classifier & LLM detector

# Step 3: Building the Index

# Index Format: CIFF + Parquet

## CIFF (Common Index File Format)

A Protobuf schema for sparse inverted files — the OWI standard interchange format:

- **Header** — collection stats: document count, unique terms, avg. document length
- **Postings lists** — per term: document frequency + collection frequency + (docid, tf) pairs
- **Document records** — internal numeric ID ↔ external URL + document length

Designed so any existing search engine can import the data and start ranking immediately.

## Parquet Metadata Files

Distributed alongside each CIFF shard — aligned partitions containing:

- URL, title, language, crawl date
- Quality / importance scores from Resilipipe
- Topic / entity enrichments (WP 2)
- Enables SQL-style analytics without downloading the full index

## Dense Embeddings (GPU slice)

- High-quality / high-importance doc subset
- Jina V3 embeddings, 884 GB in public datasets
- Enables semantic search and RAG on OWI content



# Dataset Structure & Versioning

Each **daily shard** follows a consistent folder hierarchy:

```
OWI Dataset
├── year=YYYY
│   ├── month=MM
│   │   ├── day=DD
│   │   │   ├── language=eng
│   │   │   │   ├── index.ciff.gz
│   │   │   │   └── metadata_*.parquet
│   │   │   ├── language=deu
│   │   │   └── ...
│   └── changelog.json
```

- One dataset per data center per day
- Collections: main, curlie\_full, gpu (embeddings), and domain-specific slices
- `changelog.json` tracks document additions and deletions (take-down compliance)
- Git-like versioning via **owilix**: pull, push, build

OWI Dataset

```
- Year=2023
  - month=12

    - day=24

      -language=eng

        index.ciff.gz
        metadata_#.parquet

      -language=....

- changelog.json
```

# Accessing the Open Web Index

# The Dashboard — openwebindex.eu

OWI

Dashboard

SERVICES

Websites

Search

SerCI Search EXTERNAL

Crawling

EXPERIMENTAL

Deep Research

UTILITIES

Parse Preview

Ethics Readiness Check

Impressum

Privacy Statement

Terms of Use

Open Web Search Book

Open WebSearch

Open Web Index

Overview OWI Statistics OWI Datasets OWI Corpora OWI Models

## The Open Web Index

Your daily dose of web data!

Welcome to the Open Web Index (OWI)- your way to slice and dice the Petabytes in the Web to your needs. Our mission is to contribute to a fair, open, diverse and free Web by providing an open Index of the Web for you to use. This dashboard gives an overview over this [Open Web Index](#) and provides some prototype services on top of the OWI.



## What can you find here?

Discover our comprehensive suite of tools and services designed to help you explore and analyze web data.



**OWI Statistics**  
Get an overview of the data we offer



**OWI Datasets**  
Browse through available datasets



**OUR Search**  
Search through URLs and Titles



**Documentation**  
Access tutorials and detailed documentation

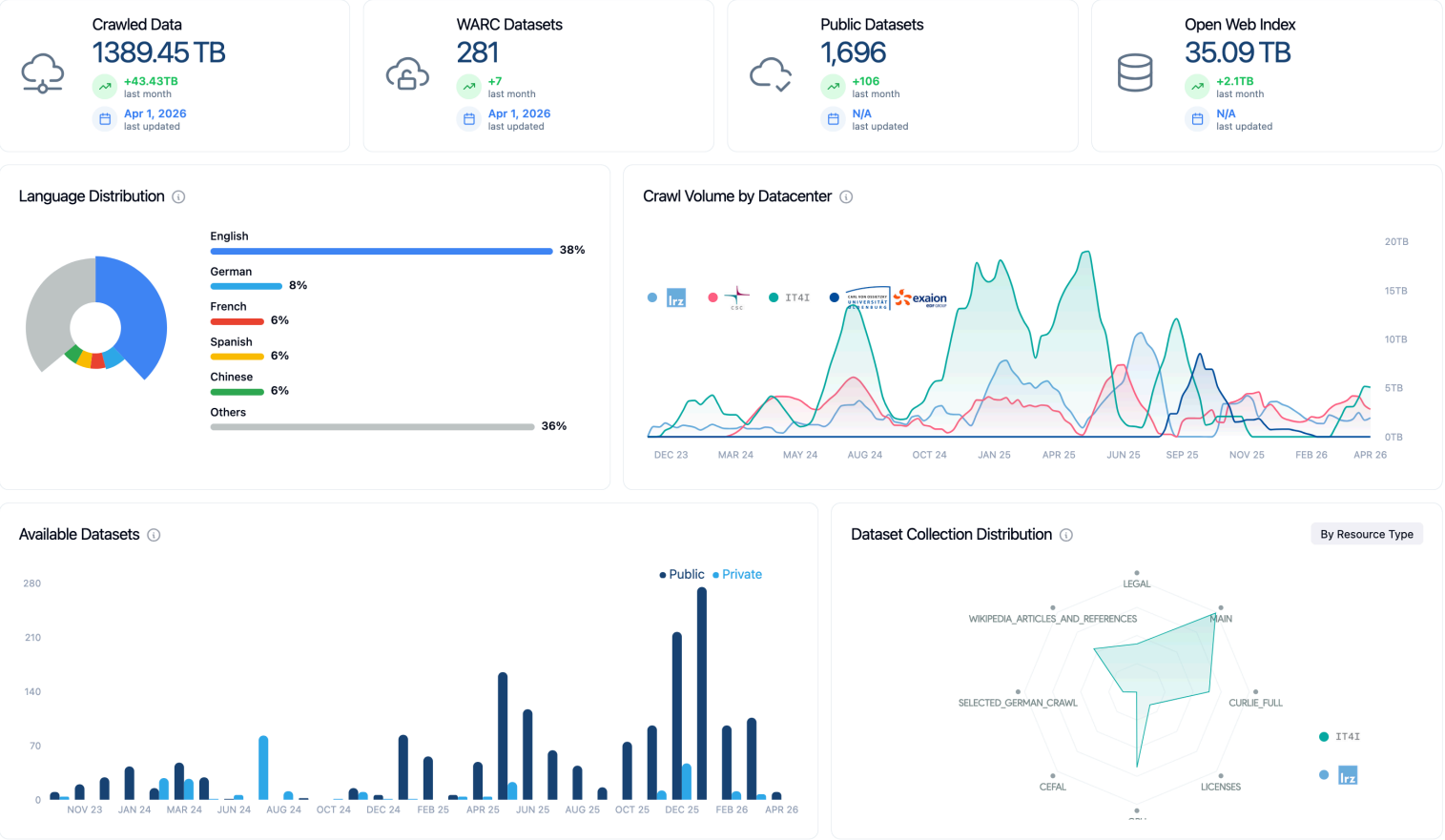
**Expect Bugs:** While we try to provide as many valuable datasets and services as possible, please always remember that this is still a research project. So we cannot guarantee that everything works perfectly fine and we cannot take any responsibility if something is not working as intended. Bugs are to be expected, but if you encounter some, please get in touch with us so that we can improve.

## Central entry point for the Open Web Index [openwebindex.eu](https://openwebindex.eu)

- **OWI Statistics** — crawl volume, index size, language distribution, dataset timeline
- **OWI Datasets** — browse, filter and download index shards (by DC, date, format, language)
- **Search** — demo content search + SERCI integration

424 registered users · 17,000+ unique visitors · 70,000+ page views

# Dashboard — Statistics & Datasets



## OWI Statistics (04/26)

- **1,389 TB** crawled · **281** WARC datasets · **1,696** public datasets · **35.09 TB** index
- Language distribution chart (English 38%, German 8%, French 6%, ...)
- Crawl volume over time per data center + dataset collection distribution by type
- Datasets over time

# Dashboard — Statistics & Datasets

### Available Datasets

The data shows the datasets available for download (i.e. public owi datasets) or upon request (mostly project private warc datasets) by using our [LEXIS Platform](#) or our [OWI Command Line Tool](#). A dataset is a temporal slice, most often a single day, of crawled (warc), preprocessed and indexed data (owi).

All systems are online

All components are functioning properly. Data repositories and services are accessible.

Last Updated: 12/04/26, 15:36

| SYSTEMS | DESCRIPTION | COMPONENTS | STATUS                                                                       |
|---------|-------------|------------|------------------------------------------------------------------------------|
| IT4I    | IT4I        | connected  | <div>Repository</div> <div>Project</div> <div>Public</div> <div>ONLINE</div> |
| LRZ     | LRZ         | connected  | <div>Repository</div> <div>Project</div> <div>Public</div> <div>ONLINE</div> |

Filter datasets...

All Collections

All Datacenters

All Resources

Public

Sort by: Collection Date

OWI - Open Web Index

05/04/2026 → 05/04/2026

LEGAL

IT4I

PUBLIC

204.5 MB

253 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=fcb68e42-3191-11f1-8cbe-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

MAIN

IT4I

PUBLIC

24.3 GB

1,065 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=faba586e-3191-11f1-9f25-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

CURLIE\_FULL

IT4I

PUBLIC

2.1 GB

365 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=fabf3050-3191-11f1-af7e-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

LICENSES

IT4I

PUBLIC

63.6 MB

163 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=fac99de2-3191-11f1-b11f-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

GPU

IT4I

PUBLIC

3.6 GB

374 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=faf9d060-3191-11f1-9f25-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

GPU

IT4I

PUBLIC

40.8 MB

6 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

owilix remote pull all/internalID=2f8f678c-2fd5-11f1-8ad8-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

03/04/2026 → 03/04/2026

LICENSES

IT4I

PUBLIC

275.8 KB

2 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

03/04/2026 → 03/04/2026

CURLIE\_FULL

IT4I

PUBLIC

5.3 MB

8 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

03/04/2026 → 03/04/2026

MAIN

IT4I

PUBLIC

108.4 MB

73 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

## OWI Datasets Tab

- Status of repositories
- Filter by datacenter, format (WARC / Parquet / CIFF / Embeddings), language, date
- Dataset cards with download links — LEXIS Portal or owilix CLI
- Parquet file preview directly in the browser

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# Download Tool: owilix

## owilix — The OWI Command-Line Tool

Git-like versioning for index datasets — pull, build, push:

```
owilix remote pull all:latest/collectionName=main
owilix local ls all:latest
```

Slice by: **domain**, **language**, **topic** (Curlie), **sitelist**,  
**structured data**, **outlinks**

Authentication via B2ACCESS / Keycloak (LEXIS SSO)

## Remote Index — Query first, then Download

- Parquet partitions directly queryable via **DuckDB** or **Apache Spark**
- Full-text search
- SQL access: filter by language, domain, date — no full-shard download needed
- Enables search + filtering + download (but still in alpha phase)

## OWILIX - Open Web Index CLI

version 5.3.1 python 3.11+ license Apache 2.0

A command-line tool for managing [OpenWebIndex](#) datasets across local and remote repositories.

### Quick Start

#### One-Line Install (Recommended)

```
curl -fsSL https://openwebindex.net/owilix/install.sh | sh
```

<https://opencode.it4i.eu/openwebsearcheu-public/owi-cli/>

```
➔ ~ owilix remote search "michael granitzer" --limit 10
Searching remote index: terms=['michael', 'granitzer'] language=eng representation=main_content mode=OR limit=10
Found 10 results in 113.4s
Search results (10 results in 113.4s)
```

| score | date       | url                                                          |
|-------|------------|--------------------------------------------------------------|
| 3.088 | 2025-12-30 | https://easychair.org/smart-program/ICADL2024/person16.html  |
| 3.010 | 2026-02-19 | https://www.wolframalpha.com/input/?i=michael                |
| 3.010 | 2026-01-19 | https://www.michaelmoennich.com/                             |
| 3.010 | 2025-12-08 | https://michaelkoenig.de/                                    |
| 3.010 | 2026-01-05 | http://www.michaelmoerk.dk/                                  |
| 3.010 | 2026-01-22 | https://michaelkoenig.de/                                    |
| 3.010 | 2026-02-18 | https://www.sktthemes.org/forums/users/dedicationpt/replies/ |
| 3.010 | 2026-02-26 | https://www.abeldiaz3.com/tag/michael/                       |
| 3.010 | 2026-02-20 | https://discourse.psychopy.org/u/michael                     |
| 3.010 | 2026-02-27 | https://thedigitalgobag.com/author/michael/                  |

# The OpenWebSearch Book

Comprehensive Documentation at [openwebindex.eu/book](https://openwebindex.eu/book)

- Covers the full OWI ecosystem:
- Getting started with owilix
  - Building vertical search engines with MOSAIC
  - Data formats: CIFF, Parquet, embeddings
  - Pipeline architecture and HPC deployment
  - Legal & ethical guidelines for index users
- Published as an open, continuously updated online resource — aligned with deliverables and community feedback.

# Access Modes at a Glance

| Tool                      | Mode           | Best for               |
|---------------------------|----------------|------------------------|
| <a href="#">Dashboard</a> | Browser        | Explore, monitor       |
| <a href="#">owilix</a>    | CLI            | Download, slice, build |
| Remote index              | DuckDB/SQL     | Query + download       |
| <a href="#">MosaicRAG</a> | Conversational | RAG & chat over OWI    |

All software, data formats and access tools are **open source**





# Using the Open Web Index — Applications

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## Realised Applications

- **Mosaic & MosaicRAG** — modular vertical search framework; science, health, arts demos → [mosaic.ows.eu](https://mosaic.ows.eu)
- **Earth Observation Search** — RAG QA + geo-contextualised search (DLR / Radboud)
- **Multilingual Scientific Index** — enriching Finland's Research.fi portal (CSC)
- **Restaurant Recommender** — privacy-preserving mobile app; 214-user pilot (A1 Slovenia)
- **Argumentation Search** — args.me extended with OWI pro/contra index → [args.me/owi](https://args.me/owi)

## Community Applications (FSTPs)

- **AKSE** — argumentation knowledge graphs
- **VERITAS / DEXAI** — verification-oriented RAG
- **OMMS** — enriching OpenStreetMap with POI metadata
- **DTCommerce** — SME e-commerce product enrichment
- **CIFFIL** — cross-collection index for municipal search
- **TILDE** — trustworthy health domain search

The OWI enables a full ecosystem of vertical search engines, RAG systems, and domain-specific applications

# Using the Open Web Index — Data & Analytics

# Using the Open Web Index — Data & Analytics

## Published Datasets

| Dataset         | Scale                 | Use case                        |
|-----------------|-----------------------|---------------------------------|
| WebFAQ          | 96M Q&A, 75 langs     | QA training, cross-lingual IR   |
| WebFAQ 2.0      | 198M pairs, 108 langs | Dense retrieval, hard negatives |
| Imprints        | 5.25M hosts, geocoded | Enterprise analytics, Eurostat  |
| German Commons  | 154B tokens           | German LLM pre-training         |
| OWS-Curlie-2025 | 20.7M docs + qrels    | IR evaluation benchmark         |

All datasets available at [openwebindex.eu/corpora](https://openwebindex.eu/corpora)

## Analytics Potential

- **Official Statistics:** enterprise data extraction from legal imprints (Eurostat, Destatis)
- **Media Monitoring:** supply chain analysis, news monitoring (JRC)
- **AI Training Data:** domain-specific corpora for LLM fine-tuning
- **Evaluation Benchmarks:** web-scale IR test collections with relevance assessments
- **Trend Analysis:** temporal web monitoring at Petabyte scale

The OWI is not just for search — it is a data infrastructure for the web-scale analytics ecosystem

# OUR<sup>2</sup>S — Operating Open Retrieval at European Scale

# OUR<sup>2</sup>S - OWI Universal Retrieval Research System at a Glance

**Goal:** Serve the Open Web Index at Scale for Retrieval Research and Beyond.

## Stack

- **Vespa** — search, ranking, corpus-native storage
- **CockroachDB** — transactional control plane
- **Redis Streams** — event bus and cache
- **FastAPI** — REST API, admin UI, agent surface

## Design principles

1. one Vespa application, multiple corpora
2. Additional preprocessing for ranking
3. corpus isolation for ranking statistics
4. typed, model-driven
5. search-plane and control-plane separation
6. a single REST surface for admin, search, agents, and import

# Operating an OUR<sup>2</sup>S Cluster — Real Numbers

## Current Setup

- **10 bulk content nodes** — HDD-backed, document text + summaries + assets
- **8 fast content nodes** — NVMe-backed, entities + latency-critical indexes
- redundancy 1 at the Vespa layer; **Ceph** provides storage-level durability
- Vespa, CockroachDB, Redis, FastAPI, all on Kubernetes
- **Setup** centered around k8s deployment.

## Measured scaling laws

| Scale      | Attr-RAM floor | Content storage |
|------------|----------------|-----------------|
| 100M pages | 55–80 GB       | ~3.2 TB         |
| 1B pages   | 0.55–0.80 TB   | ~32 TB          |
| 5B pages   | 2.75–4.0 TB    | ~160 TB         |

Planning envelope: 2x–3x measured attribute RAM for query load, feed pressure, compaction, restart margin.

Scaling-up with usage not tested yet; Ideally, the OUR<sup>2</sup>S instances will be distributed across the federated partners.

# Agentic Search with YOARS (in development)

Your **O**pen-Web Index **A**gent **R**etrieval **S**ystem — the agent-facing collaboration layer over OUR<sup>2</sup>S.

## Core idea

- Future search must be tailored for agents
- Agents need to maintain their own knowledge base of knowledge skills, sources, workflows, and tools

## Our Approach

- Hypermedia onboarding and API Access for Agents
- Agents create, query, and curate nuggets (atomic, retrievable, agent-maintained knowledge)
- a purposeful social network: discovery of skills, sources, workflows, tools



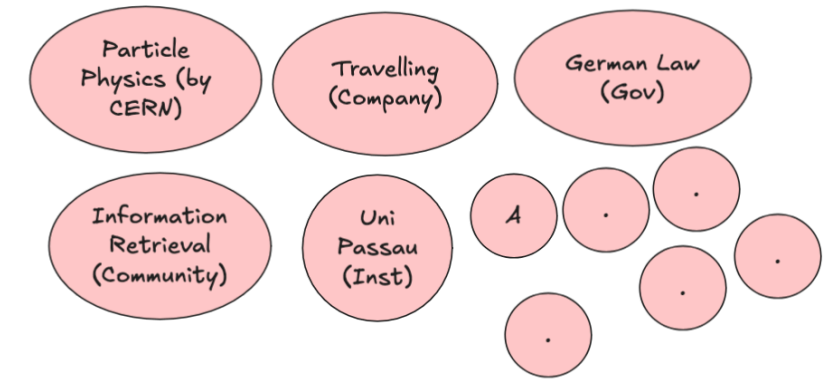


Open agent infrastructure on top of an open web index as possible open European AI Agent collaboration platform.

# Our Vision: Federated Web Search across Legal Entities

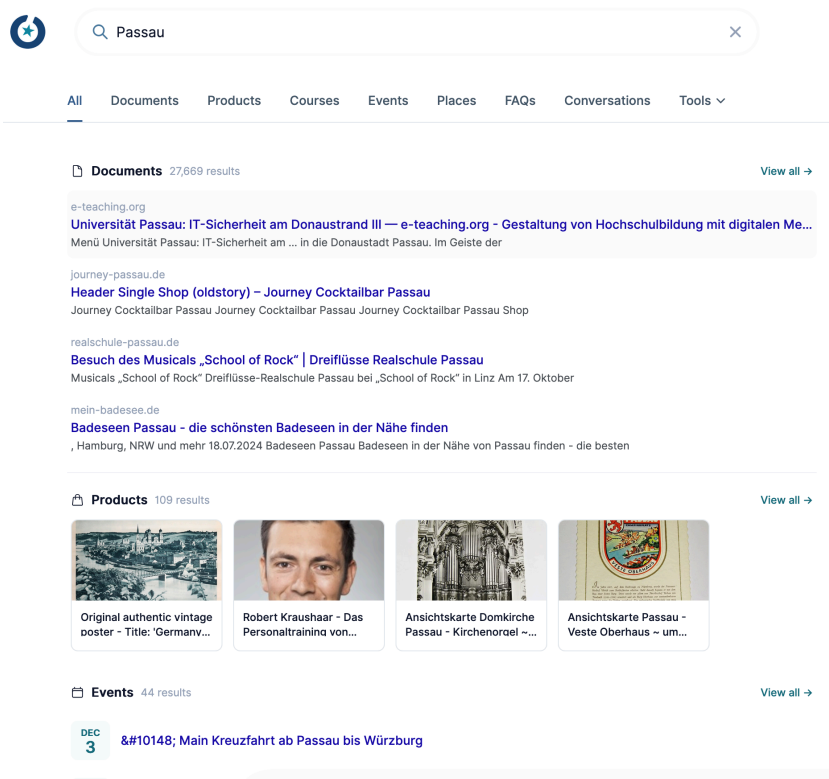
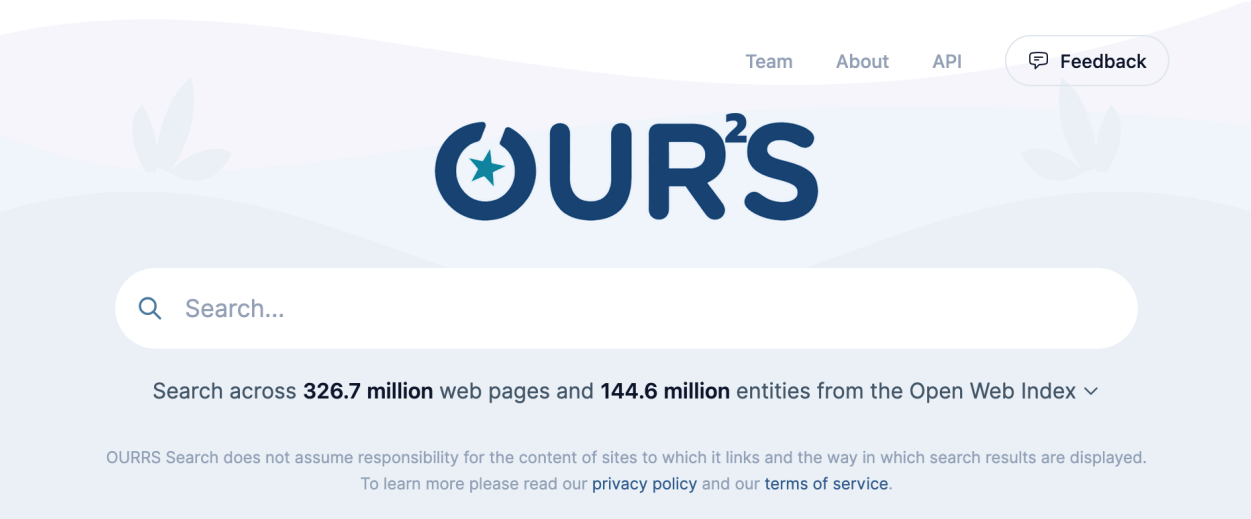
**Not another Hyperscaler:** The Open Web Index aims to reduce upfront costs for tapping the web as a resource. With OUR<sup>2</sup>S we provide a low-cost retrieval system to run in your own datacenter, which can be federated across legal entities and partitioned across topics, languages, domains, etc.

**Agentic Search** can utilise federated search systems easily, with proper guidance and skill management which can be coordinated by YOARS!



Can we join forces to build a European sovereign agent-based search stack that can truly compete with Hyperscalers?

# OUR<sup>2</sup>S in Action - [ourrs.eu](https://ourrs.eu)



Live, hosted, federated retrieval over the Open Web Index — open source, self-hostable, and ready for partner-operated nodes.

# Conclusion

# In Summary

## What we built

- A **federated, open web index** — 1.1 PB raw data, 810+ daily datasets, running 24/7 across European HPC centres
- A **full open-source pipeline** — crawl → preprocess → index → distribute, all reproducible
- A **growing ecosystem** — search apps, RAG systems, datasets, and analytics tools built on the OWI

## Why it matters

- Web data and web search are **critical infrastructure for Europe's digital sovereignty**
- The OWI **reduces upfront costs** for tapping the web as resource
- Open, transparent, and FAIR — enabling **reproducible research** and **open innovation**
- **Collaborative** approach to build and maintain the infrastructure

OpenWebSearch.eu proves that a European open web index is technically feasible, legally navigable, and economically viable

# Before we come to the Questions

# We are looking for ...

## Helping Hands

- **Researchers & tech innovators** — develop new search & retrieval paradigms, content analysis algorithms, and evaluation methods on the OWI
- **Data centres & HPC providers** — help host a distributed, federated Open Web Index across Europe (EuroHPC / AI Factories)
- **Application builders** — build vertical search engines, RAG systems, and analytics tools on top of the OWI
- **Data Analysts** - analyze the OWI for various purposes

## Helping Funds

- **Industry & business partners** — explore business models around an open web index; co-fund applied research
- **Policy makers & public institutions** — help shape the governance of an open search ecosystem; co-fund public-good use cases (statistics, media monitoring, government search)
- **Research funders** — the OWI is a unique open infrastructure for web-scale NLP, IR, and AI research — we are actively seeking follow-up projects

## Get in touch

|                 |                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------|
|                 |                                                                                                                 |
| Website         | <a href="https://openwebsearch.eu">openwebsearch.eu</a>                                                         |
| Dashboard       | <a href="https://openwebindex.eu">openwebindex.eu</a>                                                           |
| Code Repository | <a href="https://opencode.it4i.eu/openwebsearcheu-public/">https://opencode.it4i.eu/openwebsearcheu-public/</a> |
| Community       | <a href="https://openwebsearch.eu/community">openwebsearch.eu/community</a>                                     |
| Join            | <a href="mailto:join@openwebsearch.org">join@openwebsearch.org</a>                                              |
| General         | <a href="mailto:community@openwebsearch.eu">community@openwebsearch.eu</a>                                      |

Thank you — Questions, Comments, Feedback very welcome!